



WAEP Semester Two Examination, 2018

Question/Answer booklet

MATHEMATICS

METHODS

UNITS 1 AND 2 Section Two:
Calculator-assumed

SOLUTIONS

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Student number: In
figures

In words _____

Your name _____

Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper,
and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

METHODS UNITS 1 AND 2 2 CALCULATOR-ASSUMED

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination

Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Supplementary pages for the use of planning/continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

CALCULATOR-ASSUMED 3 METHODS UNITS 1 AND 2

Section Two: Calculator-assumed 65% (98 Marks)

This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9 (6 marks) (a) A sequence is defined by $u_{n+1} = u_n - 1.8$, $u_1 = 975$. Determine

(i) the value of u_{120} . (1 mark)

Solution
$u_{120} = 760.8$
Specific behaviours
✓ correct value

(ii) the sum of the first 120 terms of this sequence. (2 marks)

Solution
$u_{120} = \frac{120}{2} \times (975 + 760.8)$ $= 104\,148$
Specific behaviours
✓ uses sum formula ✓ correct sum

(b) Another sequence is defined by $u_n = 975(0.2)^{n-1}$. Determine

(i) the value of u_5 . (1 mark)

Solution
$u_5 = 1.56$
Specific behaviours
✓ correct value

(ii) the value u_n approaches as $n \rightarrow \infty$. (2 marks)

Solution
$u_\infty = \frac{975}{1 - 0.2}$ $= 1\,218.75$
Specific behaviours
✓ uses sum to infinity formula ✓ correct value

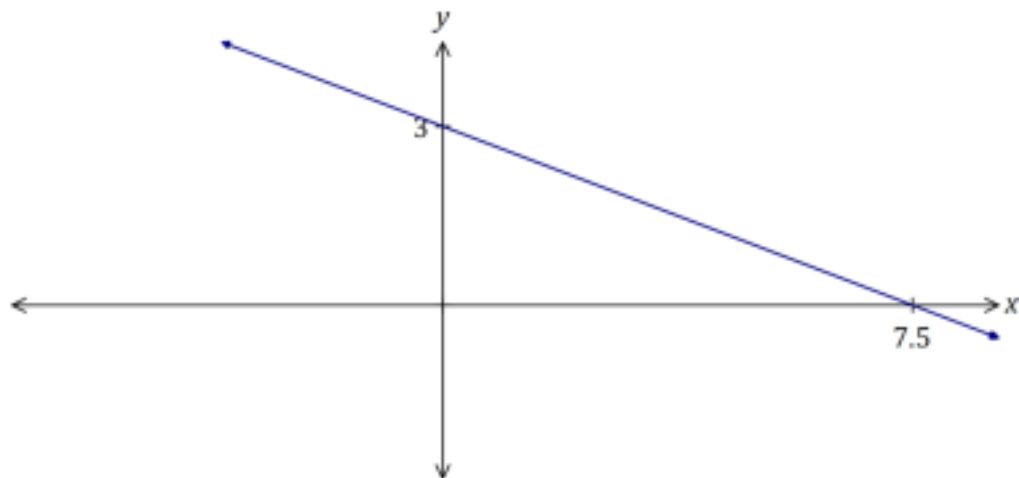
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METHODS UNITS 1 AND 2 4 CALCULATOR-ASSUMED

Question 10 (6 marks) The variables x and y are related by the equation $2x + 5y = 15$.

(a) Sketch the graph of this relationship. (2 marks)

Solution
See graph
Specific behaviours
✓ x -intercept of line ✓ y -intercept of line



(b) Express y in terms of x and briefly explain why y is a function of x . (2 marks)

Solution
$y = 3 - \frac{2}{5}x$ For each x value there is just one y value, so one-to-one relationship.
Specific behaviours
✓ correct rule ✓ indicates use of vertical line test or 'one-to-one' relationship

(c) The domain of x is restricted to $-5 \leq x < 10$. State the range of y . (2 marks)

Solution
$y = -5, y = 5, y = 10, y = -1$ $-1 < y \leq 5$
Specific behaviours
✓ correct values ✓ correct inequalities

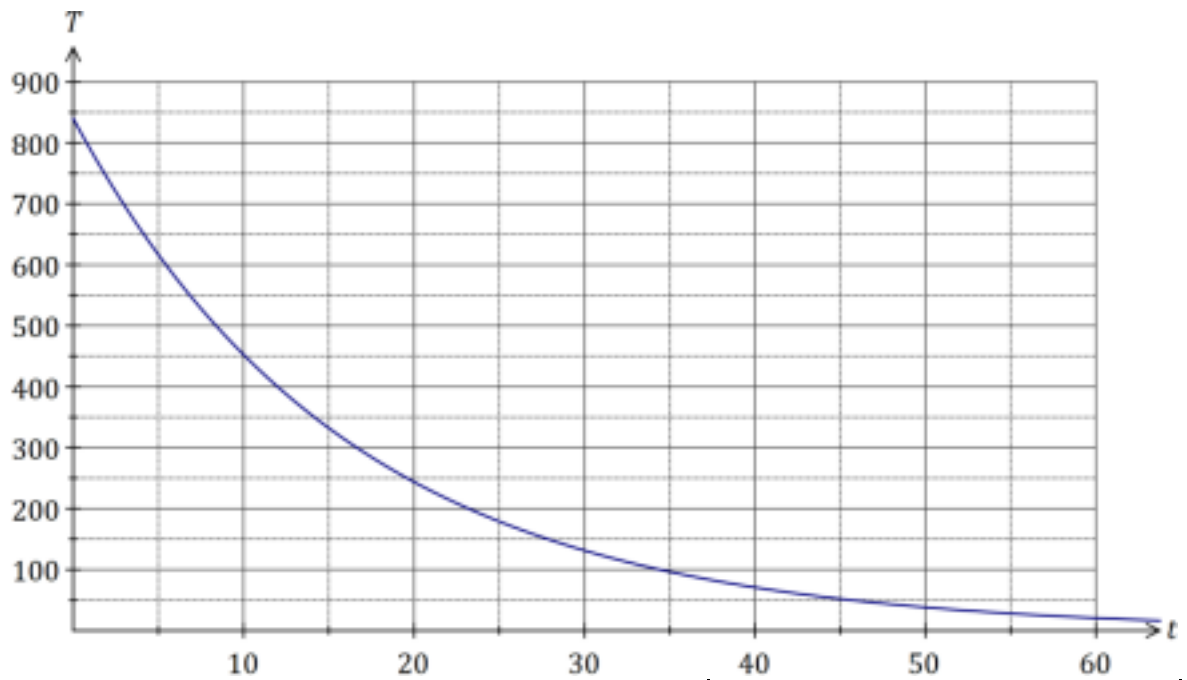
Question 11 (9 marks)

The temperature θ of a cast taken out of an oven cools according to the model $\theta = 840(0.94)^{t/5}$, where t is the time in minutes since the cast was removed from the oven. θ is measured in $^{\circ}\text{C}$.

- (a) Determine the fall in temperature of the cast during the first 5 minutes. (2 marks)

Solution
$\theta = 840(0.94)^5 \approx 616^{\circ}\text{C}$ $\Delta\theta = 840 - 616 = 224^{\circ}\text{C}$
Specific behaviours
✓ value of θ when $t = 5$ ✓ correct drop

- (b) Graph the temperature of the cast against time on the axes below. (4 marks)



Solution
See graph
Specific behaviours
✓ t-intercept ✓ close to (20, 250) and (45, 50) ✓ smooth curve throughout ✓ no upward 'hook' at RHS

(c) State the name of this type of function. (1 mark)

Solution

Exponential.
Specific behaviours
✓ correct name

(d) The temperature of the cast falls to room temperature of 15°C .

(i) Determine the time taken for the cast to reach room temperature. (1 mark)

Solution
$840(0.94)^{t/2} = 15 \Rightarrow t/2 = 65.1 \text{ m}$
Specific behaviours
✓ correct time

(ii) Comment on the usefulness of the model for large values of t . (1 mark)

Solution
For large values of t the model shows that $t/2 \rightarrow 0$ but the temperature of the cast only falls to 15°C and so model not valid for large t .
Specific behaviours
See next page ✓ states not valid, with reason

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METHODS UNITS 1 AND 2 6 CALCULATOR-ASSUMED

Question 12 (8 marks)

(a) Calculate the area of the minor segment that subtends an arc of 72° in a circle of diameter 170 cm. (2 marks)

Solution
$72^{\circ} = 2 \times 36^{\circ}$, $36^{\circ} = 170/2 = 85$ $\text{Area} = \frac{1}{2}(85)^2 \left(2 \times 36^{\circ} - \sin 2 \times 36^{\circ} \right) \approx 1\,104 \text{ cm}^2$
Specific behaviours

✓ converts angle, uses correct radius ✓ calculates area

- (b) A chord of length 26 cm subtends an angle of $22^\circ 17'$ at the centre of a circle. Calculate the radius of the circle. (2 marks)

Solution
$26 = 2r \sin \left(\frac{1}{2} \times 22^\circ 17' \right)$ $r \approx 141 \text{ cm}$
Specific behaviours
✓ substitutes into formula ✓ calculates radius

- (c) Parallelogram PQRS has side PQ = 28 cm, side QR = 19 cm and an area of 400 cm².

Determine the lengths of the diagonals of PQRS. (4 marks)

Solution
$\frac{2(28)(19) \sin \theta}{1} = 400$ $\theta = 48.75^\circ, 131.25^\circ$ $d_1 = \sqrt{28^2 + 19^2 - 2(28)(19) \cos 48.75}$ $\approx 21.1 \text{ cm}$ $d_2 = \sqrt{28^2 + 19^2 - 2(28)(19) \cos 131.25}$ $\approx 43.0 \text{ cm}$
Specific behaviours

- ✓ equation for half area
- ✓ both angles of parallelogram
- ✓ correct length of one diagonal
- ✓ second correct length

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CALCULATOR-ASSUMED 7 METHODS UNITS 1 AND 2

Question 13 (8 marks)

A mobile phone retailer classified recent sales of 625 phones by the age of customer and if the phone was bought outright or on a plan. A summary of the data is shown in the table below.

	Aged under 30	Aged 30 or over	Total
Bought outright	108	??	232
Bought on a plan	??	152	??
Total			625

(a) Determine the values of ??, ?? and ?? shown in the table. (3 marks)

Solution

$22 = 232 - 108 = 124$ $22 = 625 - 232 = 393$ $22 = 393 - 152 = 241$
Specific behaviours
✓ each correct value

(b) A recent sale is selected at random from those recorded above. Event 22 occurs if the customer was aged under 30 and event 22 occurs if the phone was bought outright.

Determine the following probabilities:

(i) $22(\overline{22})$. (1 mark)

Solution
$22(\overline{22}) = 393$ $625 (= 0.6288)$
Specific behaviours
✓ correct probability

(ii) $22(22 \cup \overline{22})$. (1 mark)

Solution
$22(22 \cup \overline{22}) = 625 - 152$ $625 = 473$ $625 (= 0.7568)$

Specific behaviours
✓ correct probability

(iii) $P(\overline{A} \cap B)$. (1 mark)

Solution
$P(\overline{A} \cap B) = \frac{124}{625} (= 0.1984)$
Specific behaviours
✓ correct probability

(iv) $P(B | \overline{A})$. (2 marks)

Solution
$P(B \overline{A}) = \frac{241}{393} (\approx 0.613)$
Specific behaviours
✓ numerator ✓ denominator

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METHODS UNITS 1 AND 2 8 CALCULATOR-ASSUMED

Question 14 (7 marks) A function is defined by $f(x) = x^4 - x^3$.

(a) Use the derivative $f'(x)$ to determine the coordinates of all stationary points of the function. (3 marks)

Solution

$$f'(x) = x^3 - 3x^2$$

$$x^3 - 3x^2 = 0 \Rightarrow x = 0, x = 3$$

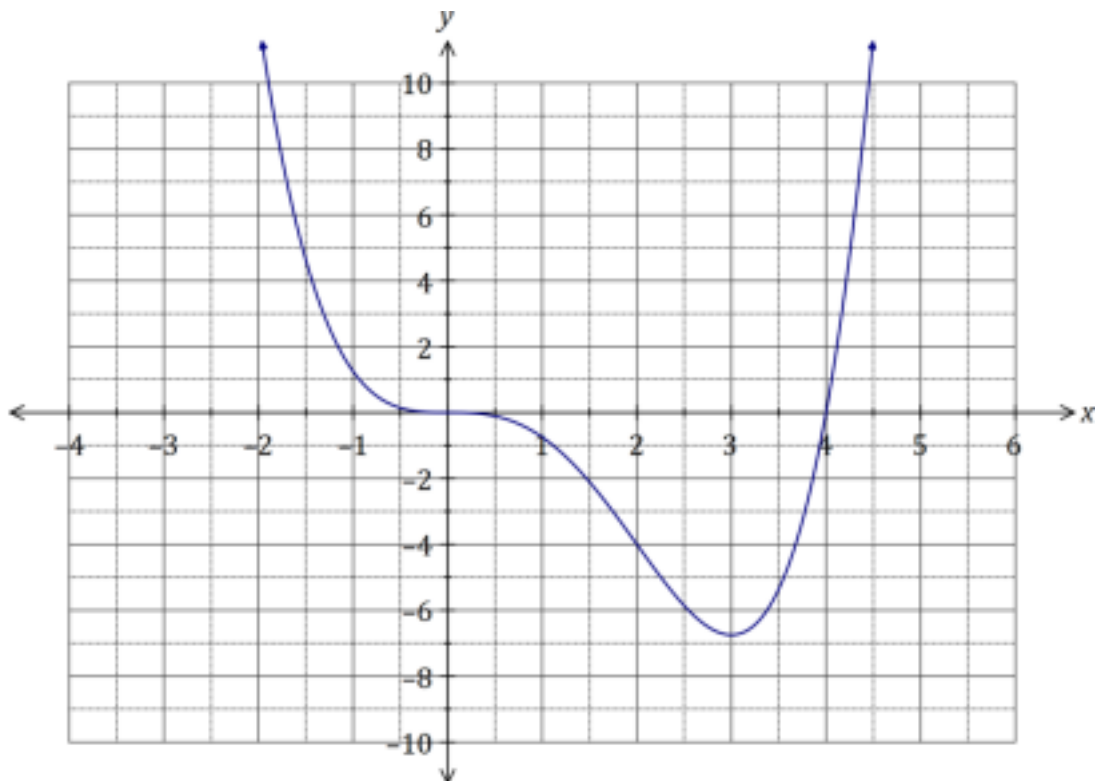
$$f(0) = 0, f(3) = -6.75$$

Stationary points at (0, 0) and (3, -6.75)

Specific behaviours

- ✓ correct derivative
- ✓ correct zeros of derivative
- ✓ correct coordinates

(b) Sketch the graph of $y = f(x)$ on the axes below. (4 marks)



Solution

See graph

Specific behaviours

- ✓ HPI at (0, 0)
- ✓ minimum at (3, -6.75)
- ✓ root at (4, 0)
- ✓ correct curvature

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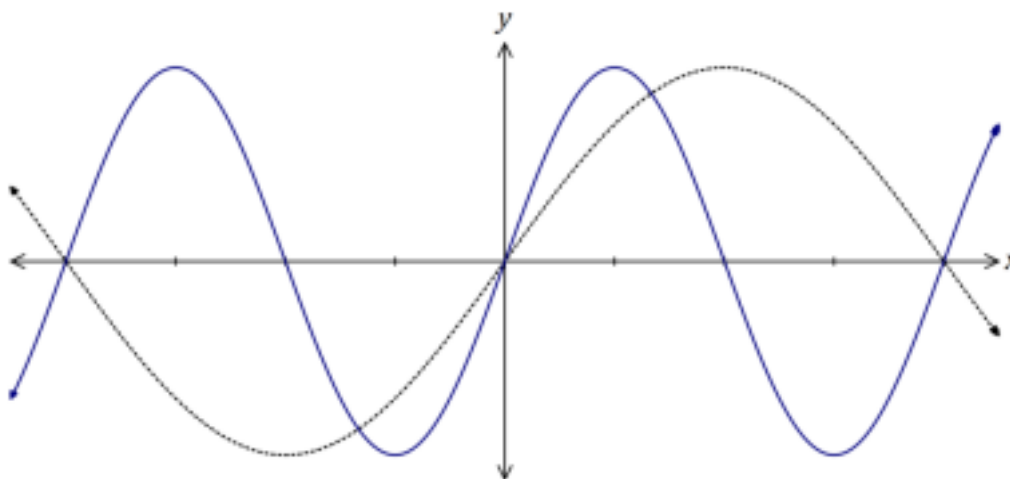
CALCULATOR-ASSUMED 9 METHODS UNITS 1 AND 2

Question 15 (7 marks) (a) The graph of $y = \sin 2x$ is shown below. On the same axes, sketch

$y = \sin$

$2x$. (2

marks)



Solution

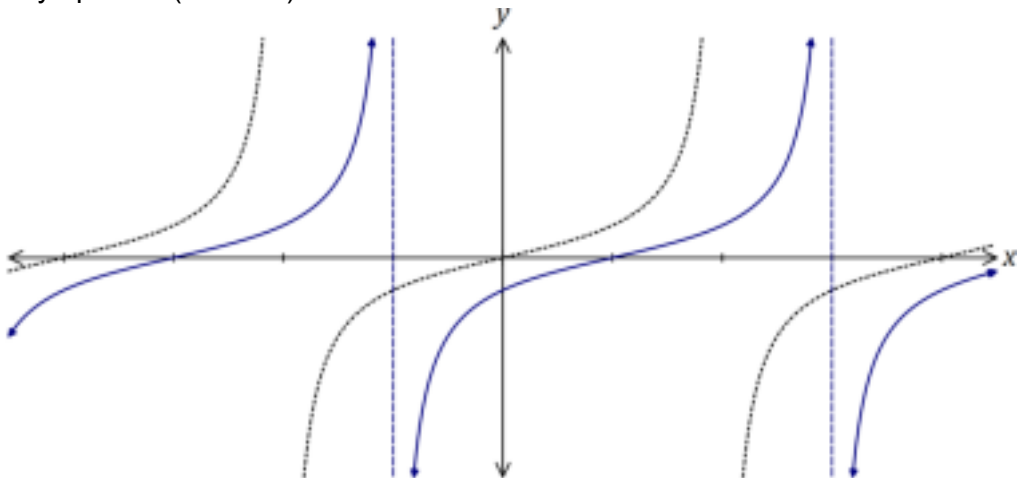
See graph

Specific behaviours

- ✓ location of roots
- ✓ correct amplitude/shape

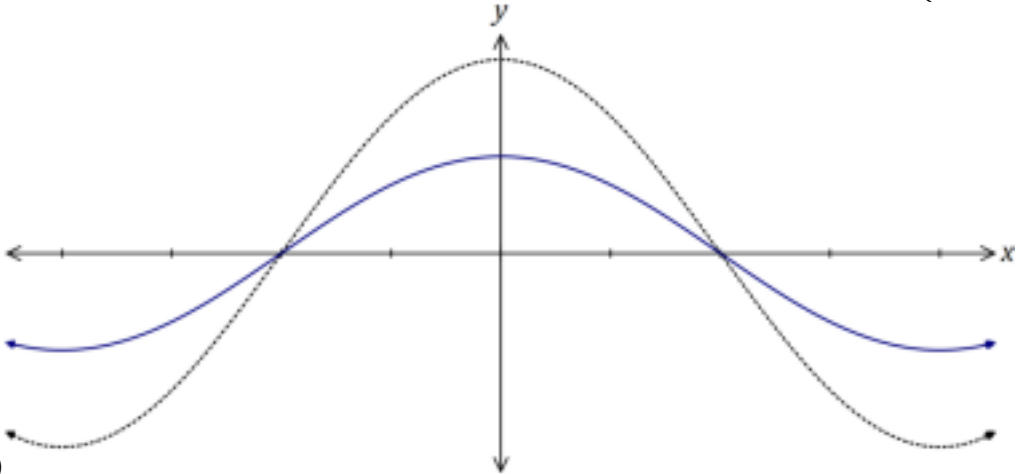
(b) The graph of $y = \tan x$ is shown below. On the same axes, sketch $y = \tan (x - \frac{\pi}{4})$, and all

its asymptotes. (3 marks)



Solution
See graph
Specific behaviours
✓ translation ✓ location of roots ✓ asymptotes

(c) The graph of $y = \cos x$ is shown below. On the same axes, sketch $y = \frac{1}{2}\sin (2x + \frac{\pi}{2})$. (2



marks)

Solution
See graph
Specific behaviours
✓ location of roots ✓ correct amplitude/shape

METHODS UNITS 1 AND 2 10 CALCULATOR-ASSUMED

Question 16 (8 marks)

A council took a random sample of 154 and 127 properties from suburbs 22 and 22 respectively. A total of 49 of the properties in the sample were in arrears with their rates, and 27 of these properties were in suburb 22. 'In arrears' means that payment of rates is overdue.

- (a) Council officers wanted to choose 5 of the properties that were in arrears. How many different selections of properties are possible? (2 marks)

Solution
${}^{49}_5 = 1\,906\,884$
Specific behaviours
✓ indicates use of ${}^{n}_r$ formula ✓ correct number

- (b) Determine the probability that one randomly chosen property from the sample

- (i) is in suburb 22 and is in arrears. (2 marks)

Solution
$\frac{27}{154 + 127} = \frac{27}{281} \approx 0.078$
Specific behaviours
✓ numerator ✓ denominator

- (ii) is not in arrears given that it is in suburb 22. (1 mark)

Solution

$P(A) = 154 - 27$ $154 = 127$ $154 (\approx 0.825)$
Specific behaviours
✓ correct probability

- (c) Justifying your answer with conditional probabilities, comment on whether being in arrears with rates is independent of the suburb the property is in. (3 marks)

Solution
$P(A B) = 27$ $154 \approx 17.5\%$ $P(A C) = 22$ $127 \approx 17.3\%$ Hence being in arrears is independent of suburb, as conditional probabilities are very similar.
Specific behaviours
✓ calculates $P(A B)$ ✓ calculates $P(A C)$ ✓ correct conclusion

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CALCULATOR-ASSUMED 11 METHODS UNITS 1 AND 2

Question 17 (8 marks)

Two water containers, initially empty, are being filled with water. The amount of water added to container A each minute follows an arithmetic sequence, with 3 mL poured in during the first minute and 6 mL poured in during the second minute. The amount of water added to container B each minute follows a geometric sequence, with 2 mL poured in during the first minute and 2.2 mL poured in during the second minute.

- (a) The amount of water poured into container A during the n^{th} minute is given by $a + (n-1)d$. State the value of the constants a and d . (2 marks)

Solution

$V_0 = 2$ $V_1 = 2 \cdot 2 = 1.1$
Specific behaviours
✓ value of V_0 ✓ value of V_1

(b) Determine the total amount of water in container V_0 at the end of the 25th minute.

(2 marks)

Solution
$V_{25} = 2(1 - 1.1^{25})$ $1 - 1.1$ $= 197 \text{ mL}$
Specific behaviours
✓ uses sum formula ✓ correct amount

(c) How long does it take to fill container V_0 with 360 mL of water? (2 marks)

Solution
V_0 $2(2(3) + (V_0 - 1)(3)) =$ $360 \quad V_0 = 15 \text{ minutes}$
Specific behaviours
✓ uses sum formula ✓ correct time

(d) Container 22 first holds more water than container 22 at the end of minute 22.

(i) Determine the value of 22. (1 mark)

Solution
22 = 59
Specific behaviours
✓ correct value

(ii) State, to the nearest mL, how much more water 22 contains than 22 at this time.

✓ correct value

Solution
5516 – 5310 = 206 mL SN078-122-4 (1 mark)
Specific behaviours

See next page

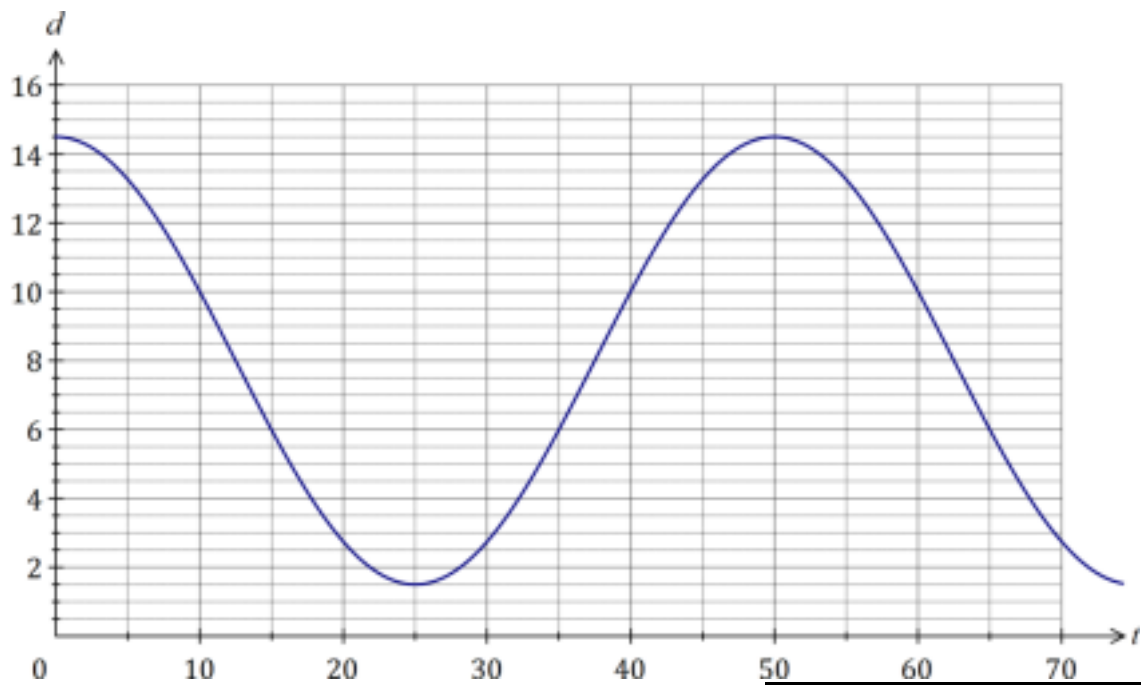
METHODS UNITS 1 AND 2 12 CALCULATOR-ASSUMED

Question 18 (8 marks)

The height, h metres, above level ground of a seat on a steadily rotating Ferris wheel 22 seconds after observations began was given by

$$h = 6.5 \cos \left(\frac{2222}{25} \right) + 8, 22 \geq 0.$$

(a) Draw the graph of the height of the seat against time on the axes below. (4 marks)



Solution
See graph
Specific behaviours
✓ correct t -intercept ✓ accurate curve near maximum ✓ accurate curve near minimum ✓ smooth curve throughout

(b) How long did the Ferris wheel take to complete one revolution? (1 mark)

Solution
50 seconds
Specific behaviours
✓ correct time

(c) At what time, when the seat was rising, did it first reach a height of 11 metres? (1 mark)

Solution
$t = 41.3 \text{ s}$
Specific behaviours
✓ time that rounds to 41 s

(d) Determine the change in height of the seat between $t = 130$ and $t = 131$, giving your answer rounded to the nearest cm. (2 marks)

Solution
$h(130) = 2.74, h(131) = 3.26$ $\Delta h = 3.26 - 2.74 = 0.52 \text{ m}$
Specific behaviours
✓ determines both heights ✓ states difference to nearest cm

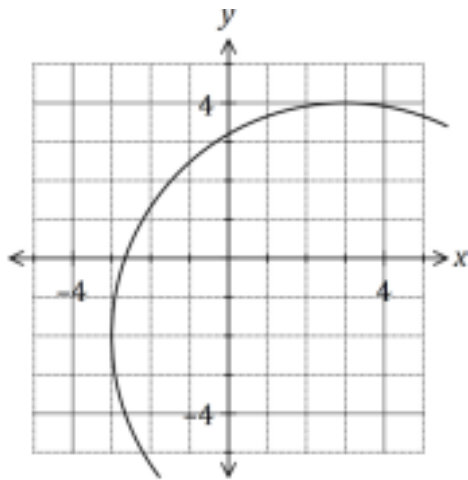
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CALCULATOR-ASSUMED 13 METHODS UNITS 1 AND 2

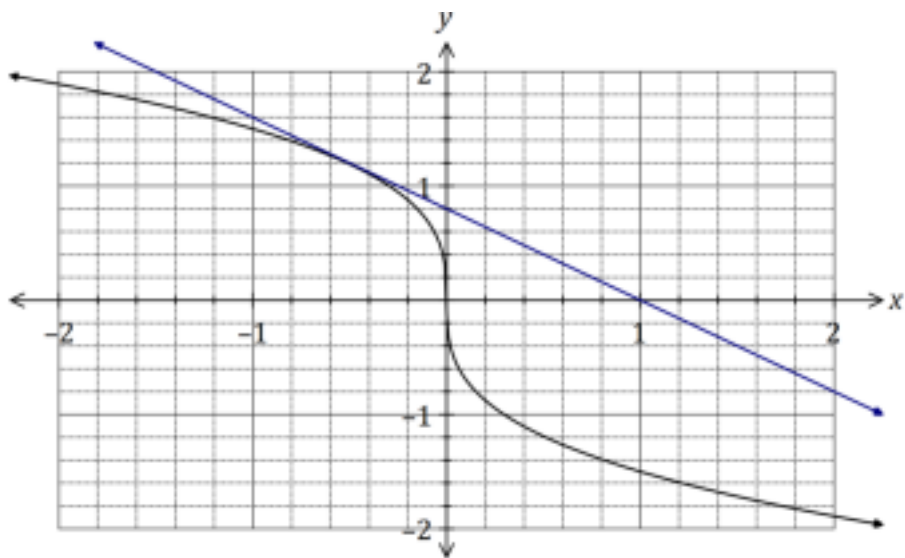
Question 19 (7 marks)

(a) Part of the circle $x^2 + y^2 = px + qy + r$ is shown below. Determine the values of the constants p , q and r . (4 marks)

Solution
$(x - 3)^2 + (y + 2)^2 = 6^2$ $x^2 + y^2 = 6x - 4y + 23$ $p = 6, q = -4, r = 23$
Specific behaviours
✓ circle in factored form ✓ correct radius and centre ✓ expands into required form ✓ correct values of p , q and r



(b)
The
graph
of a power function $y = x^2(2x)$ is shown below.



Solution
See graph
Specific behaviours
✓ tangent drawn with ruler

(i) Draw the tangent to the curve when $x = -0.5$. (1 mark)

(ii) Use the tangent to estimate $f'(-0.5)$. (2 marks)

Solution
$f'(-0.5) \approx -1.62 \approx -0.8$
Specific behaviours
✓ uses gradient of tangent ✓ value $-1 \leq f' \leq -0.7$

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METHODS UNITS 1 AND 2 14 CALCULATOR-ASSUMED

Question 20 (8 marks)

A diagnostic test for a disease has a 97% chance of giving the correct outcome and it is known that 0.5% of all sheep on a station have the disease. It can be assumed that the correct outcome of the test is independent of whether a sheep has the disease.

(a) A sheep is randomly selected for the test from those on the station. Determine the probability that

(i) the sheep has the disease, but the test indicates that it does not. (2 marks)

Solution
$P = 0.005 \times 0.03 = 0.00015$ $0.00015 \times 20\,000 = 3$
Specific behaviours
✓ indicates multiplication of correct probabilities ✓ correct probability

(ii) the sheep actually has the disease if the test indicates that it does. (4 marks)

Solution
$P(D \cap T) = 0.005 \times 0.97 = 0.00485$ $P(\bar{D} \cap T) = 0.995 \times 0.03 = 0.02985$ $P(T) = 0.00485 + 0.02985 = 0.0347$ $P(D T) = \frac{0.00485}{0.0347} \approx 0.140$
Specific behaviours

- ✓ calculates $P(A \cap B)$
- ✓ calculates $P(\bar{A} \cap B)$
- ✓ indicates $P(A)$
- ✓ correct probability

(b) Two sheep are randomly selected for the test from those on the station. Determine the probability that just one of the sheep is diagnosed correctly. (2 marks)

Solution
$P(A_1 \cap A_2) = 0.97 \times 0.03 = 0.0291$ $P(\bar{A}_1 \cap A_2) = 0.03 \times 0.97 = 0.0291$ $P(A) = 0.0291 + 0.0291 = 0.0582$
Specific behaviours
✓ indicates correct method ✓ correct probability

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CALCULATOR-ASSUMED 15 METHODS UNITS 1 AND 2

Question 21 (8 marks)

A pyramid with a rectangular base of length l and width w has perpendicular height h . The length of the base is three times its width and the sum of the width, length and height is 180 cm.

(a) Calculate the length, height and volume of the pyramid when $w = 15$ cm. (2 marks)

Solution

$l = 3 \times 15 = 45, h = 180 - 15 - 45 = 120$ $V = \frac{1}{3}(15 \times 45) \times 120 = 27\,000 \text{ cm}^3$
Specific behaviours
✓ correct length and height ✓ correct volume

(b) Show that the volume of the pyramid is given by $V = 180x^2 - 4x^3$. (2 marks)

Solution
$l = 3x, h = 180 - x - 3x = 180 - 4x$ $V = \frac{1}{3}(x \times 3x)(180 - 4x)$ $= 180x^2 - 4x^3$
Specific behaviours
✓ expressions for length and height ✓ substitutes width, length and height correctly

(c) Use calculus to determine the maximum volume of the pyramid and state the dimensions required to achieve this volume. (4 marks)

Solution

$V = 180x^2 - 4x^3$ $V' = 360x - 12x^2$ $360x - 12x^2 = 0 \Rightarrow x = 0, 30$ $V_{\text{max}} = 180(30)^2 - 4(30)^3 = 54\,000 \text{ cm}^3$ $x = 30 \text{ cm}, y = 90 \text{ cm}, h = 60 \text{ cm}$
Specific behaviours
✓ correct derivative using given variables ✓ solves derivative equal to zero ✓ correct maximum volume ✓ correct dimensions

SN078-122-4 **End of questions**

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Supplementary page

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CALCULATOR-ASSUMED 17 METHODS UNITS 1 AND 2

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METHODS UNITS 1 AND 2 18 CALCULATOR-ASSUMED

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CALCULATOR-ASSUMED 19 METHODS UNITS 1 AND 2

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Supplementary page

Question number: _____

